Biotechnology and its Applications

Biotechnology essentially deals with industrial scale production of biopharmaceuticals and biologicals, using genetically modified microbes, fungi, plants and animals.

Where it is applied

- therapeutics and diagnostics
- genetically modified crops for agriculture, and processed food
- bioremediation, waste treatment and energy production

Three critical research areas

- providing the best catalyst, in the form of an improved organism, usually a microbe or a pure enzyme
- creating optimal conditions through engineering for that catalyst to act
- downstream processing technologies to purify the protein or organic compound

Three options for more food

- this chapter looks at how human beings have used biotechnology to improve the quality of human life, especially in food production and health
- agro-chemical based agriculture is the first of the three options that can be thought of for increasing food production

In this chapter

10.2 Tissue culture
10.4 Genetically modified organisms
10.5 Bt cotton
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10.10 Gene therapy
10.12 Transgenic animals

10.1 Three options for more food

The other two options

- organic agriculture
- genetically engineered crop-based agriculture

Why the Green Revolution was not enough

- it succeeded in tripling the food supply, yet that was not enough to feed the growing human population
- increased yields came partly from improved crop varieties, but mainly from better management practices and the use of agrochemicals, that is fertilisers and pesticides
- for farmers in the developing world agrochemicals are often too expensive, and further increases in yield with existing varieties are not possible by conventional breeding

10.2 Tissue culture

Totipotency and micro-propagation

- as traditional breeding techniques failed to keep pace with demand and to provide sufficiently fast and efficient systems for crop improvement, another technology, tissue culture, was developed
- scientists learnt during the 1950s that whole plants could be regenerated from explants, that is any part of a plant taken out and grown in a test tube under sterile conditions in a special nutrient medium, and this capacity to generate a whole plant from any cell or explant is called totipotency
- the nutrient medium must provide a carbon source such as sucrose, and also inorganic salts, vitamins, amino acids and growth regulators such as auxins and cytokinins
- by these methods a large number of plants can be propagated in very short durations, and producing thousands of plants this way is called micro-propagation

Somaclones, and virus-free plants

- each of these plants is genetically identical to the original plant from which it was grown, that is, they are somaclones
- many important food plants such as tomato, banana and apple have been produced on a commercial scale by this method, and a visit to a tissue culture laboratory is worth making to see the process
- another important application is the recovery of healthy plants from diseased ones, since even in a virus infected plant the apical and axillary meristem is free of virus, so the meristem can be removed and grown in vitro to obtain virus-free plants
- scientists have succeeded in culturing meristems of banana, sugarcane and potato

Naked protoplasts

- scientists have isolated single cells from plants and, after digesting their cell walls, have been able to isolate naked protoplasts, surrounded by plasma membranes
- isolated protoplasts from two different varieties of plant, each having a desirable character, can be fused to get hybrid protoplasts
- these can be grown further to form a new plant, and such hybrids are called somatic hybrids

10.3 Protoplasts and somatic hybrids

Somatic hybridisation

- the process of fusing protoplasts is called somatic hybridisation
- a protoplast of tomato fused with that of potato and then grown gives a new hybrid plant combining tomato and potato characteristics, the pomato
- this has been achieved, but the plant did not have all the desired combination of characteristics for commercial use
- the use of genetically modified crops is a possible solution to obtaining maximum yield while minimising the fertilisers and chemicals whose harmful effects fall on the environment

10.4 Genetically modified organisms

What genetic modification has done

- plants, bacteria, fungi and animals whose genes have been altered by manipulation are called genetically modified organisms (GMO)
- it has made crops more tolerant to abiotic stresses, that is cold, drought, salt and heat, and reduced reliance on chemical pesticides through pest-resistant crops
- it has helped reduce post harvest losses, and increased the efficiency of mineral usage by plants, which prevents early exhaustion of the fertility of the soil
- it has enhanced the nutritional value of food, as in golden rice, that is Vitamin A enriched rice, and it has been used to create tailor-made plants supplying alternative resources to industries in the form of starches, fuels and pharmaceuticals

10.5 Bt cotton

The bio-pesticide

- one application in agriculture is the production of pest resistant plants, which could decrease the amount of pesticide used
- Bt toxin is produced by a bacterium called Bacillus thuringiensis, Bt for short, and the Bt toxin gene has been cloned from the bacteria and expressed in plants to give resistance to insects without the need for insecticides, in effect creating a bio-pesticide
- examples are Bt cotton, Bt corn, rice, tomato, potato and soyabean

How the toxin works

- some strains of Bacillus thuringiensis produce proteins that kill certain insects such as lepidopterans, that is tobacco budworm and armyworm, coleopterans, that is beetles, and dipterans, that is flies and mosquitoes
- the bacterium forms protein crystals during a particular phase of its growth, and these crystals contain a toxic insecticidal protein
- the Bt toxin protein exists as an inactive protoxin, so it does not kill the Bacillus, but once an insect ingests the inactive toxin the alkaline pH of the gut solubilises the crystals and converts it into an active form
- the activated toxin binds to the surface of the midgut epithelial cells and creates pores that cause cell swelling and lysis, and eventually the death of the insect

Cotton boll destroyed by bollworms, on the one side

A fully mature cotton boll, on the other

The difference a Bt gene makes in the field

Figure 10.A Cotton boll: (a) destroyed by bollworms; (b) a fully mature cotton boll

10.6 Cry genes

Choosing the gene for the pest

- specific Bt toxin genes were isolated from Bacillus thuringiensis and incorporated into several crop plants such as cotton
- the choice of gene depends on the crop and the targeted pest, as most Bt toxins are insect-group specific
- the toxin is coded by a gene crylAc named cry, and there are a number of them, so that the proteins encoded by the genes crylAc and crylIAb control the cotton bollworms and that of crylAb controls corn borer

10.7 Pest resistant plants and RNA interference

The nematode problem

- several nematodes parasitise a wide variety of plants and animals, human beings included
- the nematode Meloidegyne incognitia infects the roots of tobacco plants and causes a great reduction in yield
- a novel strategy was adopted to prevent this infestation, based on the process of RNA interference (RNAi)

What RNA interference is

- RNAi takes place in all eukaryotic organisms as a method of cellular defense
- the method involves the silencing of a specific mRNA by a complementary dsRNA molecule that binds to it and prevents its translation
- the source of this complementary RNA could be an infection by viruses having RNA genomes, or mobile genetic elements, that is transposons, which replicate through an RNA intermediate

Roots of a typical control plant, open to infestation

Transgenic plant roots five days after deliberate infection

The transgenic roots protected, the control roots not

Figure 10.B Host plant-generated dsRNA triggers protection against nematode infestation: (a) Roots of a typical control plants; (b) transgenic plant roots 5 days after deliberate infection of nematode but protected through novel mechanism.

Silencing the nematode's own mRNA

- using Agrobacterium vectors, nematode-specific genes were introduced into the host plant
- the DNA was introduced in such a way that it produced both sense and anti-sense RNA in the host cells
- these two RNAs, being complementary to each other, formed a double stranded dsRNA that initiated RNAi and so silenced the specific mRNA of the nematode
- the consequence was that the parasite could not survive in a transgenic host expressing the specific interfering RNA, so the transgenic plant protected itself from the parasite

10.8 Applications in medicine

Recombinant therapeutics

- the recombinant DNA technological processes have made an immense impact on healthcare, by enabling the mass production of safe and more effective therapeutic drugs
- the recombinant therapeutics do not induce the unwanted immunological responses common with similar products isolated from non-human sources
- at present about 30 recombinant therapeutics have been approved for human use the world over, and 12 of these are presently being marketed in India

Pro-insulin, the form first synthesised

C peptide, the extra stretch it carries

Chain A and chain B, held by disulphide bridges

Mature insulin, after the C peptide is removed

Figure 10.C Maturation of pro-insulin into insulin (simplified)

10.9 Genetically engineered insulin

Why a bacterial source was wanted

- management of adult-onset diabetes is possible by taking insulin at regular time intervals, and without enough human insulin one would have to isolate and use insulin from other animals, which raises the question of whether it is as effective and whether it would elicit an immune response
- insulin used for diabetes was earlier extracted from the pancreas of slaughtered cattle and pigs, and insulin from an animal source caused some patients to develop allergy or other reactions to the foreign protein
- if a bacterium were available that could make human insulin the whole process becomes simple, since a large quantity of the bacteria can easily be grown and as much insulin made as is needed
- whether insulin can be orally administered to diabetic people, and why, is worth thinking through

How it was assembled

- insulin consists of two short polypeptide chains, chain A and chain B, linked together by disulphide bridges
- in mammals, humans included, insulin is synthesised as a pro-hormone, which like a pro-enzyme needs to be processed before it becomes a fully mature and functional hormone, and it contains an extra stretch called the C peptide, which is not present in mature insulin and is removed during maturation
- the main challenge for producing insulin by rDNA techniques was getting it assembled into a mature form
- in 1983 Eli Lilly, an American company, prepared two DNA sequences corresponding to chains A and B of human insulin, introduced them into plasmids of E. coli to produce the chains separately, and then extracted and combined them by creating disulfide bonds to form human insulin

10.10 Gene therapy

Correcting a gene defect

- gene therapy is a collection of methods that allows correction of a gene defect diagnosed in a child or embryo, genes being inserted into a person's cells and tissues to treat a disease
- correction involves delivering a normal gene into the individual or embryo, to take over the function of and compensate for the non-functional gene
- the first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency, an enzyme crucial for the immune system to function, the disorder being caused by deletion of the gene for adenosine deaminase
- in some children ADA deficiency can be cured by bone marrow transplantation and in others treated by enzyme replacement therapy, in which functional ADA is given by injection, but neither approach is completely curative

The lymphocyte route

- as a first step towards gene therapy, lymphocytes from the blood of the patient are grown in a culture outside the body
- a functional ADA cDNA, using a retroviral vector, is then introduced into these lymphocytes, which are subsequently returned to the patient
- as these cells are not immortal the patient requires periodic infusion of such genetically engineered lymphocytes
- if the gene isolated from marrow cells producing ADA were introduced into cells at early embryonic stages, it could be a permanent cure

10.11 Molecular diagnosis

Detecting a pathogen early

- for effective treatment of a disease, early diagnosis and an understanding of its pathophysiology matter greatly, and conventional methods such as serum and urine analysis do not allow early detection
- recombinant DNA technology, polymerase chain reaction (PCR) and enzyme linked immuno-sorbent assay (ELISA) are some of the techniques that serve the purpose
- the presence of a pathogen is normally suspected only once it has produced a disease symptom, by which time its concentration in the body is already very high, whereas a very low concentration of a bacterium or virus can be detected by amplifying its nucleic acid by PCR
- PCR is now routinely used to detect HIV in suspected AIDS patients, to detect mutations in genes in suspected cancer patients, and as a powerful techqnique to identify many other genetic disorders

Probes and ELISA

- a single stranded DNA or RNA tagged with a radioactive molecule, the probe, is allowed to hybridise to its complementary DNA in a clone of cells, followed by detection using autoradiography
- the clone having the mutated gene will not appear on the photographic film, because the probe will not have complementarity with the mutated gene
- ELISA is based on the principle of antigen-antibody interaction, so infection by a pathogen can be detected by the presence of antigens such as proteins and glycoproteins, or by detecting the antibodies synthesised against the pathogen

10.12 Transgenic animals

What they are, and two reasons for them

- animals whose DNA has been manipulated to possess and express an extra, that is foreign, gene are known as transgenic animals, and transgenic rats, rabbits, pigs, sheep, cows and fish have been produced, although over 95 per cent of all existing transgenic animals are mice
- normal physiology and development: they can be specifically designed to allow the study of how genes are regulated and how they affect the normal functions of the body and its development, as with the complex factors involved in growth such as insulin-like growth factor, where introducing genes from other species that alter the formation of the factor and studying the biological effects yields information about its role in the body
- study of disease: many are designed to increase our understanding of how genes contribute to the development of disease, and are specially made to serve as models for human diseases such as cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's, so that investigation of new treatments is made possible

10.13 More uses of transgenic animals

Biological products

- medicines required to treat certain human diseases can contain biological products, but such products are often expensive to make
- transgenic animals that produce useful biological products can be created by introducing the portion of DNA, that is the genes, which codes for a particular product, such as the human protein α-1-antitrypsin used to treat emphysema, and similar attempts are being made for the treatment of phenylketonuria (PKU) and cystic fibrosis
- in 1997 the first transgenic cow, Rosie, produced human protein-enriched milk at 2.4 grams per litre, the milk containing human alpha-lactalbumin and being nutritionally a more balanced product for human babies than natural cow-milk

Testing safety

- vaccine safety: transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans, and are being used to test the safety of the polio vaccine, so that if this proves successful and reliable they could replace the use of monkeys to test batches of the vaccine
- chemical safety testing, known as toxicity or safety testing, follows the same procedure as that used for testing the toxicity of drugs
- transgenic animals are made that carry genes which make them more sensitive to toxic substances than non-transgenic animals, they are then exposed to those substances and the effects are studied, and toxicity testing in such animals allows results to be obtained in less time

10.14 Ethical issues

Regulation, and why it is needed

- the manipulation of living organisms by the human race cannot go on any further without regulation, and some ethical standards are required to evaluate the morality of all human activities that might help or harm living organisms
- beyond the morality of such issues, the biological significance also matters, since genetic modification of organisms can have unpredicatable results when such organisms are introduced into the ecosystem
- the Indian Government has therefore set up organisations such as GEAC, the Genetic Engineering Approval Committee, which makes decisions on the validity of GM research and the safety of introducing GM-organisms for public services
- the modification and usage of living organisms for public services, as food and medicine sources for example, has also created problems with the patents granted for the same, and there is growing public anger that certain companies are being granted patents for products and technologies that make use of genetic materials, plants and other biological resources long identified, developed and used by farmers and indigenous people of a specific region or country

10.15 Patents and biopiracy

The Basmati case

- rice is an important food grain whose presence goes back thousands of years in Asia's agricultural history, and there are an estimated 200,000 varieties of rice in India alone, the diversity of rice in India being one of the richest in the world
- Basmati rice is distinct for its unique aroma and flavour, 27 documented varieties of Basmati are grown in India, and there is reference to Basmati in ancient texts, folklore and poetry, as it has been grown for centuries
- in 1997 an American company got patent rights on Basmati rice through the US Patent and Trademark Office, which allowed it to sell a new variety of Basmati in the US and abroad, a variety actually derived from Indian farmer's varieties, Indian Basmati having been crossed with semi-dwarf varieties and claimed as an invention or a novelty
- the patent extends to functional equivalents, implying that other people selling Basmati rice could be restricted by it, and several attempts have also been made to patent uses, products and processes based on Indian traditional herbal medicines such as turmeric and neem, so unless these patent applications are countered immediately other countries or individuals may encash on this rich legacy

Biopiracy

- biopiracy is the term used for the use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned, and without compensatory payment
- most industrialised nations are rich financially but poor in biodiversity and traditional knowledge, while the developing and underdeveloped world is rich in both
- traditional knowledge related to bio-resources can be exploited to develop modern applications, and can also save time, effort and expenditure during commercialisation

The response

- there has been growing realisation of the injustice, the inadequate compensation and the benefit sharing between developed and developing countries
- some nations are therefore developing laws to prevent such unauthorised exploitation of their bio-resources and traditional knowledge
- the Indian Parliament has recently cleared the second amendment of the Indian Patents Bill, which takes such issues into consideration, including patent terms of emergency provisions, research, and development initiatives

10.16 Summary

Plants

- biotechnology has given humans several useful products by using microbes, plants, animals and their metabolic machinery, and the techniques of tissue culture and somatic hybridisation offer vast potential for manipulating plants in vitro to produce new varieties
- recombinant DNA technology has made it possible to engineer microbes, plants and animals so that they have novel capabilities, and genetically modified organisms have been created by using methods other than natural ones to transfer one or more genes from one organism to another, generally by techniques such as recombinant DNA technology
- GM plants have been useful in increasing crop yields, reducing post-harvest losses and making crops more tolerant of stresses, and there are several GM crop plants with improved nutritional value of foods and reduced reliance on chemical pesticides through pest resistance

Medicine and animals

- recombinant DNA technological processes have made an immense impact on healthcare by enabling mass production of safe and more effective therapeutics, and since the recombinant therapeutics are identical to human proteins they do not induce unwanted immunological responses and are free from the risk of infection observed with similar products isolated from non-human sources, human insulin being made in bacteria yet absolutely identical in structure to the natural molecule
- transgenic animals are also used to understand how genes contribute to the development of a disease, by serving as models for human diseases such as cancer, cystic fibrosis, rheumatoid arthritis and Alzheimer's
- gene therapy is the insertion of genes into an individual's cells and tissues to treat diseases, especially hereditary diseases, by replacing a defective mutant allele with a functional one, or by gene targeting which involves gene amplification, and viruses that attack their hosts and introduce their genetic material into the host cell as part of their replication cycle are used as vectors to transfer healthy genes, or more recently portions of genes
- the current interest in the manipulation of microbes, plants and animals has raised serious ethical questions

10.17 Exercises

From this chapter

1. Which part of the plant is best suited for making virus-free plants and why?
2. What is the major advantage of producing plants by micropropagation?
3. Find out what the various components of the medium used for propagation of an explant in vitro are?
4. Crystals of Bt toxin produced by some bacteria do not kill the bacteria themselves because – (a) bacteria are resistant to the toxin

10.18 Exercises, continued

From this chapter, continued

1. (b) toxin is immature; (c) toxin is inactive; (d) bacteria encloses toxin in a special sac.
2. What are transgenic bacteria? Illustrate using any one example.
3. Compare and contrast the advantages and disadvantages of production of genetically modified crops.
4. What are Cry proteins? Name an organism that produce it. How has man exploited this protein to his benefit?
5. What is gene therapy? Illustrate using the example of adenosine deaminase (ADA) deficiency.
6. Digrammatically represent the experimental steps in cloning and expressing an human gene (say the gene for growth hormone) into a bacterium like E. coli ?
7. Can you suggest a method to remove oil (hydrocarbon) from seeds based on your understanding of rDNA technology and chemistry of oil?
8. Find out from internet what is golden rice.
9. Does our blood have proteases and nucleases?
10. Consult internet and find out how to make orally active protein pharmaceutical. What is the major problem to be encountered?